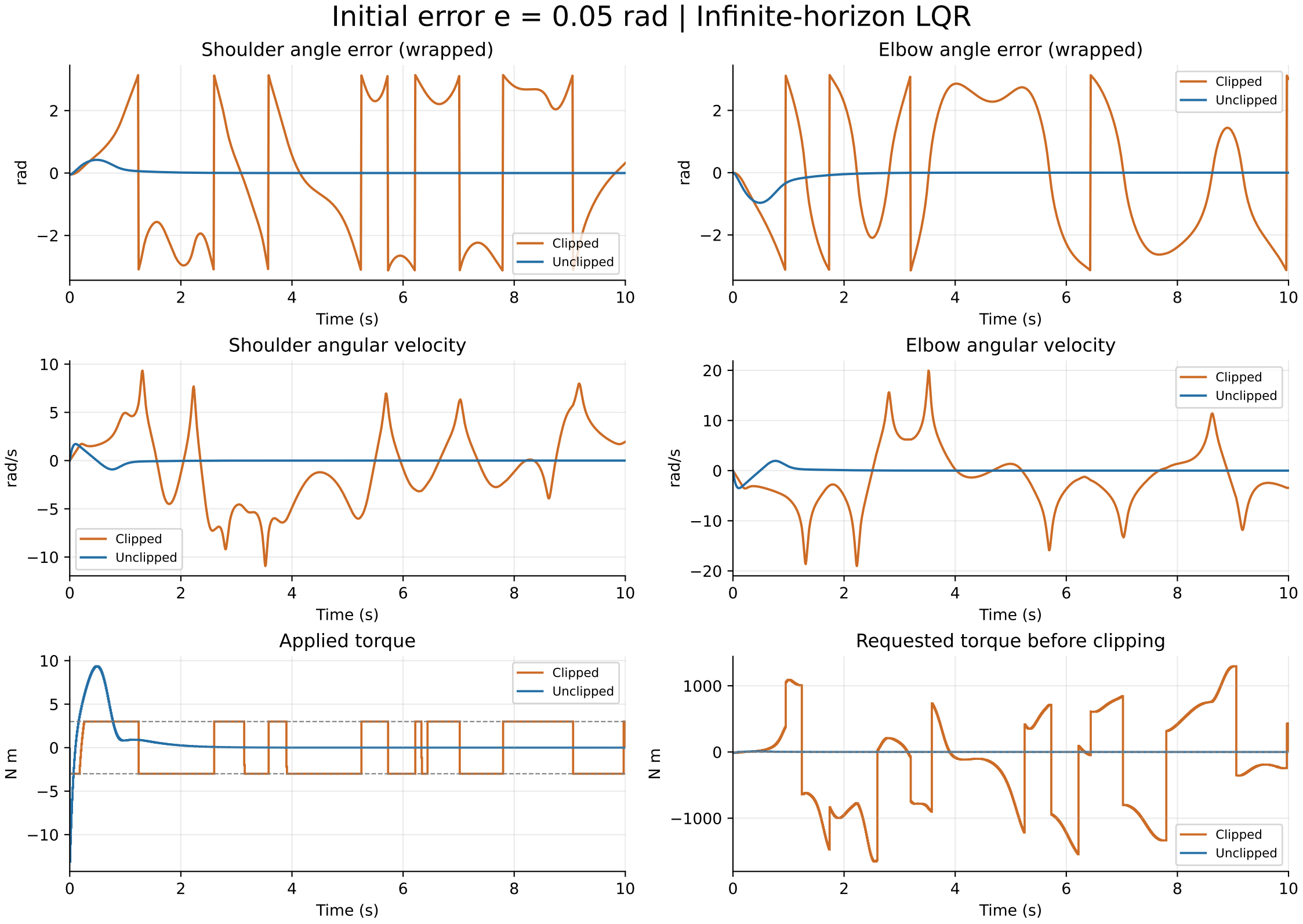
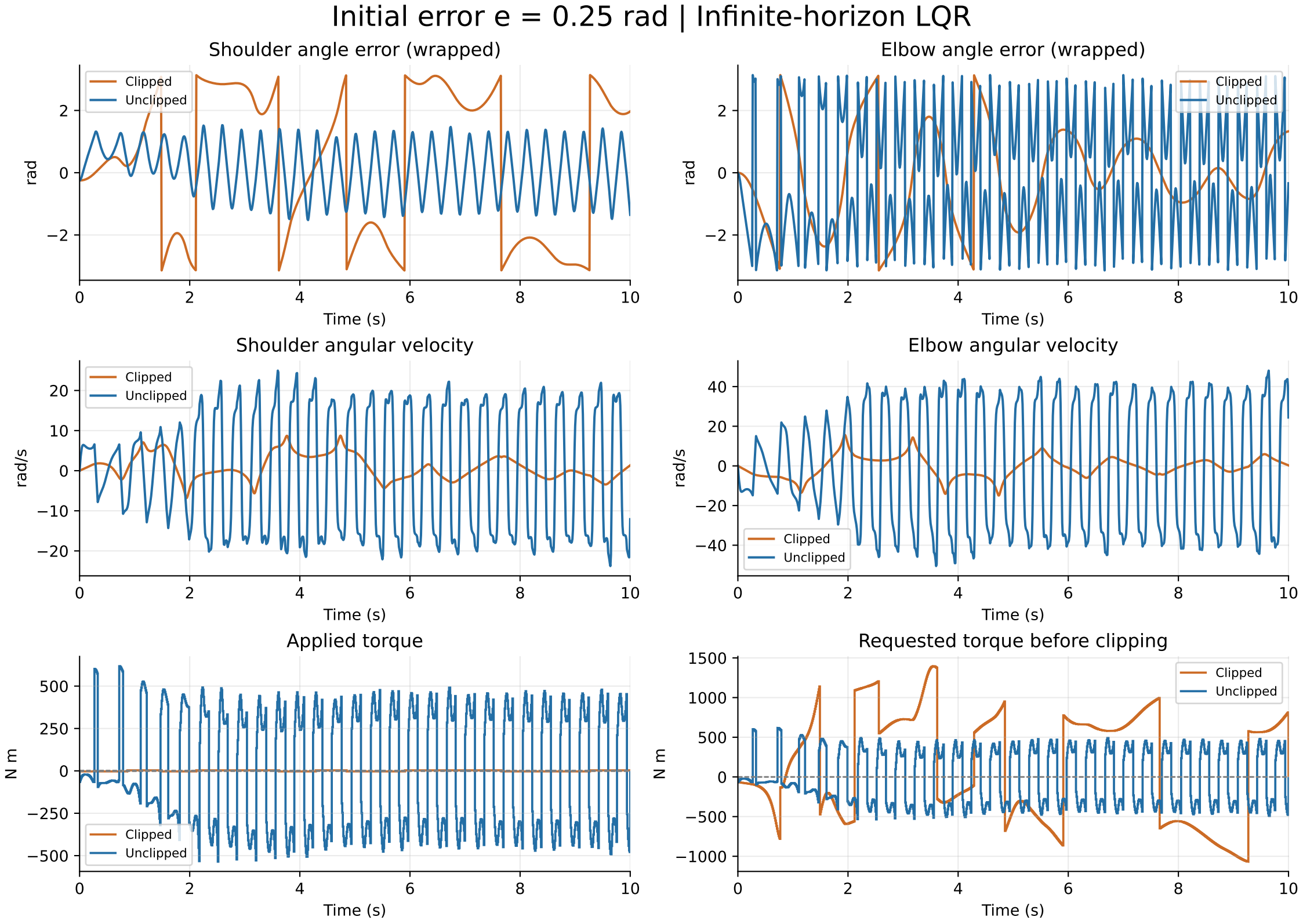
# Acrobot | Infinite-horizon LQR

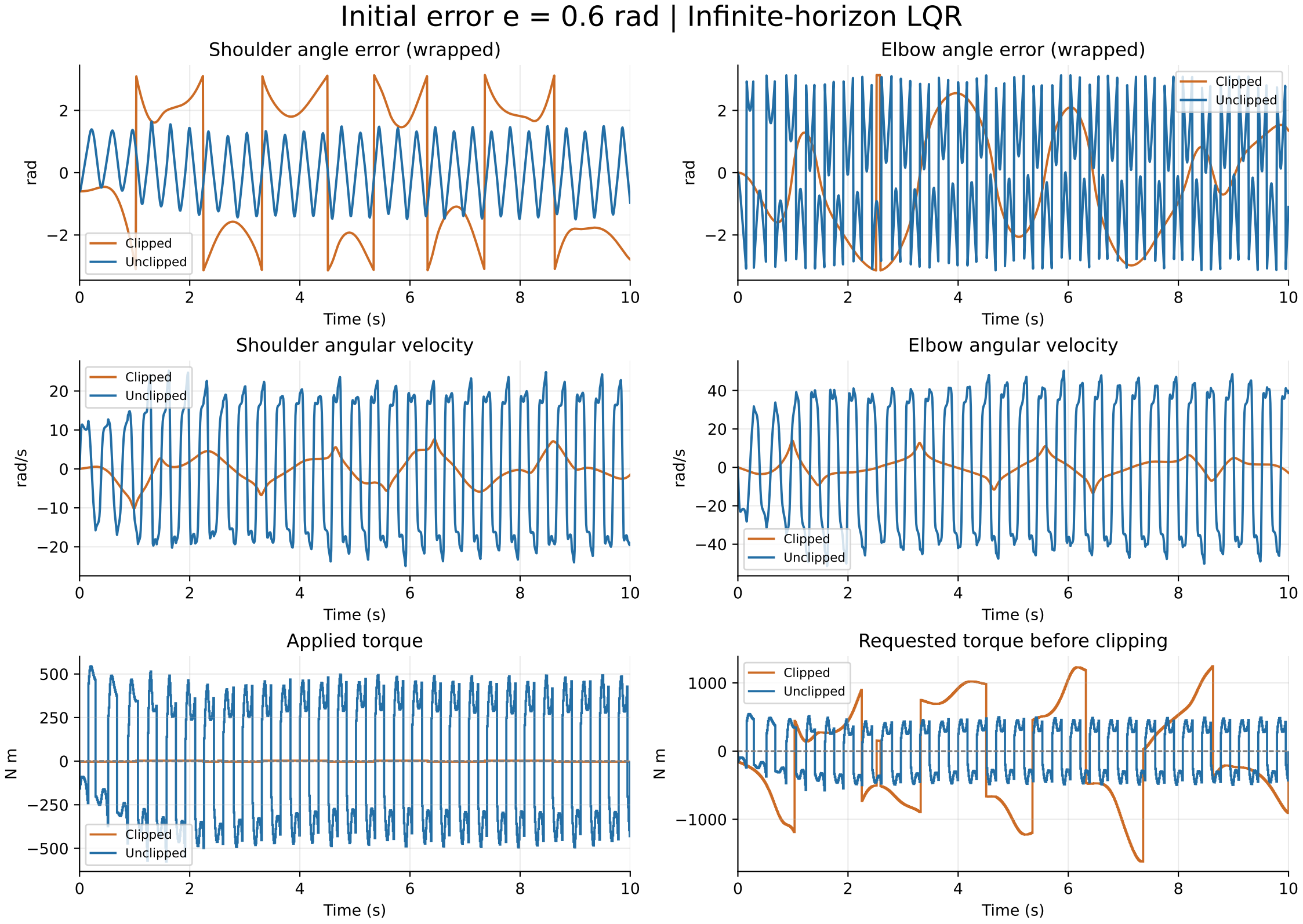
Six nonlinear Drake simulations | 10 s | control period 0.01 s
Initial state: [pi - e, 0, 0, 0], e = 0.05, 0.25, 0.6 rad.
Clipped torque: +/-3 N m. Unclipped: no torque limit.
Q = diag(10, 10, 1, 1); R = 1; u = -K error (angles wrapped to [-pi, pi)).
One constant gain is used for every run; saturation does not redesign K.
K = [-262.131925 -105.190703 -112.642345  -53.383978]

Local closed-loop eigenvalues (discrete time):
Euler design A - BK: 0.902955+0.000000j, 0.958624+0.000000j, 0.977823+0.000205j, 0.977823-0.000205j
Exact zero-order hold: 0.900450+0.000000j, 0.957031+0.000000j, 0.979234+0.000000j, 0.977844+0.000000j
Spectral radii: design 0.977823; exact 0.979234.
These poles apply near upright, where clipping is inactive.
Saturated nonlinear motion has no single fixed closed-loop pole set.

| e (rad) | Clipping | Peak \|u\| (N m) | Clipped steps | Final angle norm (rad) | Final speed norm (rad/s) |
|---|---|---|---|---|---|
| 0.05 | On | 3.000 | 989 | 3.03 | 3.95 |
| 0.05 | Off | 13.107 | 0 | 3.31e-09 | 6.91e-09 |
| 0.25 | On | 3.000 | 999 | 2.38 | 1.37 |
| 0.25 | Off | 617.364 | 0 | 1.39 | 27.2 |
| 0.6 | On | 3.000 | 1000 | 3.1 | 3.26 |
| 0.6 | Off | 576.331 | 0 | 1.46 | 43.1 |


Initial error e = 0.05 rad | Infinite-horizon LQR
Shoulder angle error (wrapped)
Elbow angle error (wrapped)
Shoulder angular velocity
Elbow angular velocity
Applied torque
Requested torque before clipping
rad
rad/s
N m
Time (s)
Clipped
Unclipped


Initial error e = 0.25 rad | Infinite-horizon LQR
Shoulder angle error (wrapped)
Elbow angle error (wrapped)
Shoulder angular velocity
Elbow angular velocity
Applied torque
Requested torque before clipping
Clipped
Unclipped
rad
rad/s
N m
Time (s)


Initial error e = 0.6 rad | Infinite-horizon LQR
Shoulder angle error (wrapped)
Elbow angle error (wrapped)
Shoulder angular velocity
Elbow angular velocity
Applied torque
Requested torque before clipping
rad
rad/s
N m
Time (s)
Clipped
Unclipped